

WASP-135b

R-1914

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-135_260630 · created 2026-08-05T23:52:02+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461221.8722 +/- 0.0079 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.024
Transit depth (Rp/Rs)^2	3.06 +/- 0.84 [%]
Orbital Inclination (inc)	82.06 +/- 1.71
Airmass coefficient 1 (a1)	0.996 +/- 0.029
Airmass coefficient 2 (a2)	0.005 +/- 0.018
Scatter in the residuals of the lightcurve fit is	2.92 %
Best Comparison Star	#3 - [322, 436]
Optimal Method	PSF photometry
Transit Duration (day)	0.071 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.175 ± 0.024 vs 0.139 (deviation 1.43σ)
Duration fit vs published	0.071 d vs 0.0687 d (deviation 0.15σ)
Mid-time O–C	-18.2 min
Depth SNR	7.0
Residual scatter	2.92 %

Light curve

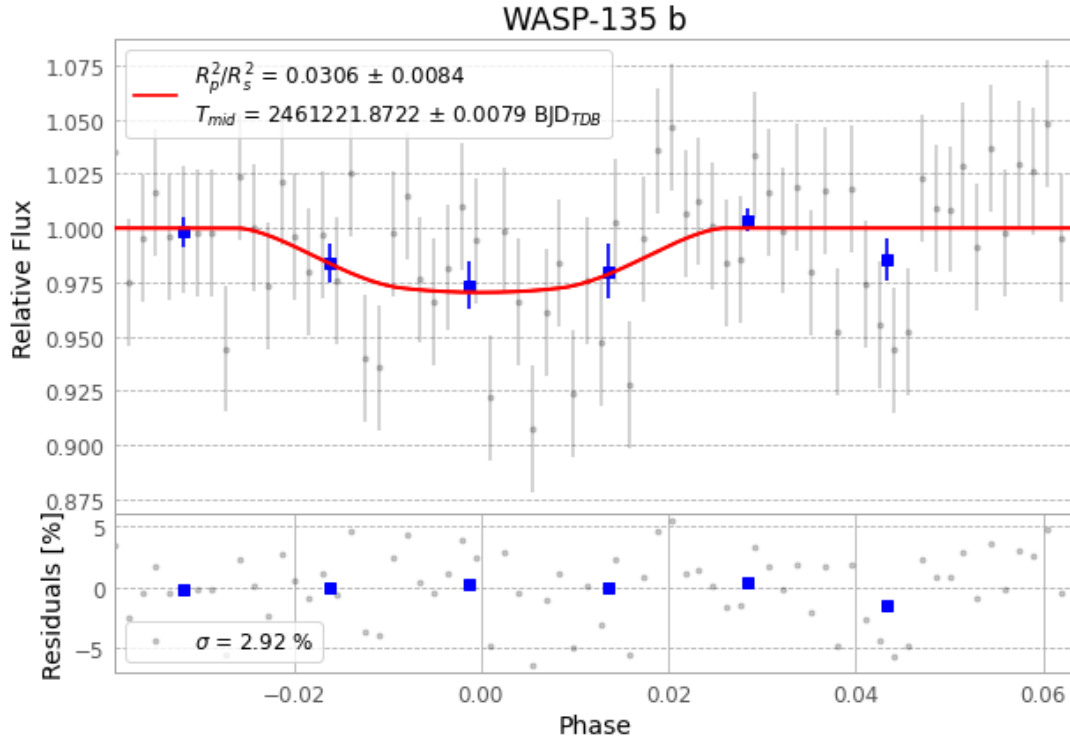


Figure 1 — FinalLightCurve_WASP-135 b_2026-06-30.png (SHA-256 0b91ce318af1ed99...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [267, 186], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5e9a4e59dbabf6fd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

70 FITS frames (46 MB), WASP-135260630073028.FITS → WASP-135260630105715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-260630.log (4599c0a85003...), FinalLightCurve_WASP-135 b_2026-06-30.png (0b91ce318af1...), FinalLightCurve_WASP-135 b_2026-06-30.csv (44a6a3db10f7...)

Compute resources

Wall time 127.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-05 23:52Z.